

Practical WS11: Planning Questions

Name: **SOLUTIONS** ()

Class: **26S**

Date:

[2011/P2/Q7]

When a wire is stretched between two points A and B and is made to vibrate, stationary waves can be set up on the wire. A possible shape (mode of vibration) for the stationary waves is shown in Fig. 1.

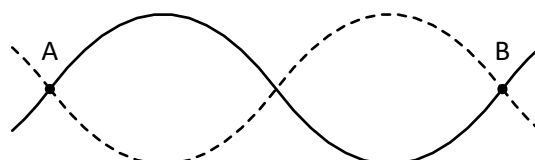


Fig. 1

The frequency of vibration of the wire in this mode of vibration is thought to depend on the tension in the wire, the length of the wire and the mass per unit length of the wire.

With all other variables kept constant, the relation between the frequency f of this mode of vibration of the wire and the mass per unit length μ of the wire is

$$f = k\mu^n$$

where k and n are constants.

You are provided with an oscillator that can vibrate the wire over a large range of unknown frequencies. You may also use any of the other equipment usually found in a physics laboratory.

Design an experiment using different wires to determine the value of n .

You should draw a labelled diagram to show the arrangement of your apparatus. In your account you should pay particular attention to

- (a) the equipment you would use,
- (b) the procedure to be followed,
- (c) the control of variables,
- (d) how the frequency of vibration of the wire would be measured,
- (e) any precautions that would be taken to improve the accuracy of the experiment.

[12]

Problem Definition

Independent variable:	Mass per unit length μ of wire
Dependent variable:	Frequency f of stationary wave
Control variables:	i. Length L of the wire ii. Tension F_T of the wire

Experimental Set-up

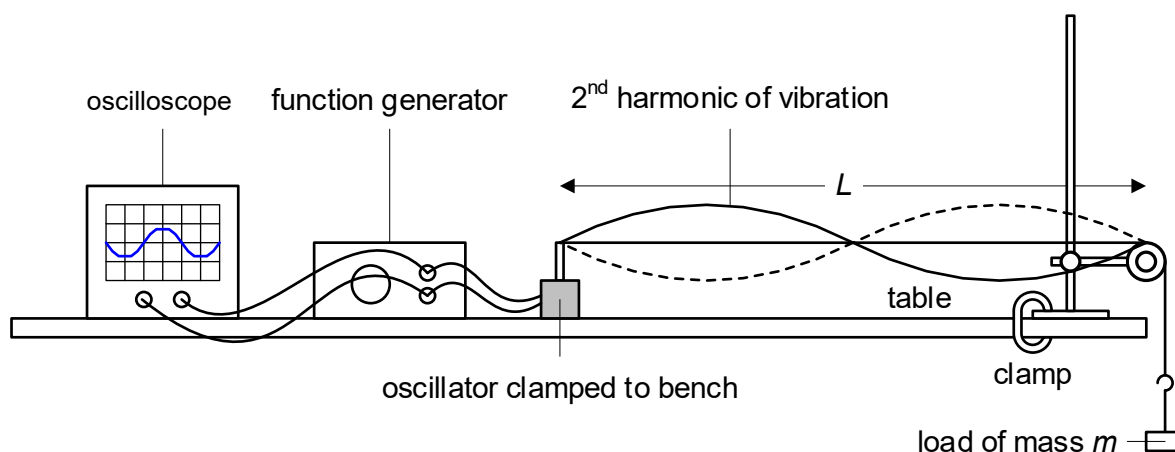


Fig. 2

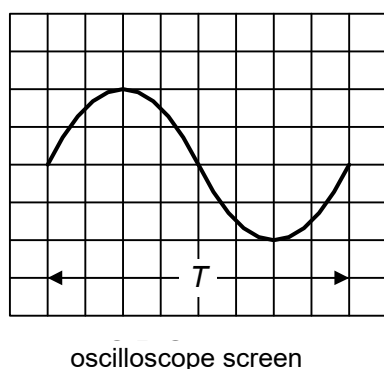


Fig. 3a

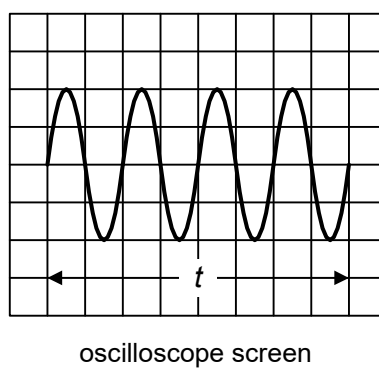


Fig. 3b



Fig. 4

Procedure

Determining the mass per unit length μ of wire [heading not required in report]

1. Measure the mass m of the wire using an [electronic balance](#) and its length L using a [metre rule](#).
2. Determine mass per unit length of wire using $\mu = \frac{m}{L}$.

Determining the frequency of stationary wave [heading not required in report]

3. Set up the apparatus as shown in Fig. 2.
4. Slowly increase the frequency of the function generator starting from 0 Hz until a stationary wave of 2 loops with a large amplitude (Fig. 1) is formed.
5. Measure the number of horizontal divisions for one complete waveform and multiply it by the time-base (time/div) to obtain the period T .
6. Determine the resonant frequency using $f = \frac{1}{T}$.
7. Repeat steps 1 to 6 to obtain a total of 8 sets of values of μ and f by using wires of different densities or thickness.
8. The tension F_T in the wire is kept constant by using load of the same mass.
9. The length L of wire is kept constant by checking with a metre rule.

Reliability

10. Adjust the time-base (time/div) of the oscilloscope to produce a large complete waveform on the screen to reduce the percentage uncertainty in the measurement of T (see Fig. 3a above).

[Alternatively, determine the time t for N number of complete waveforms (Fig. 3b) such that period $T = t/N$.]

11. **Before collecting any data, check that the load allows all wires to produce the stationary wave shown in Fig. 1.**

Analysis

Since $f = k\mu^n$, $\lg f = \lg k + n \lg \mu$.

Plot a graph of $\lg f$ against $\lg \mu$.

The value of n is obtained from the gradient of the graph.

Safety Precautions

Wear protective eyewear to prevent injuries to the eyes in case the wire snaps.

Wear protective footwear to prevent injuries to the feet due to falling load.

Secure the oscillator and the retort stand to the bench with G-clamps to prevent them from toppling and causing injuries.

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Suggested mark Scheme	
Detailed Diagram	[max 2]
Workable and secure arrangement of oscillator, wire and pulley.	0
Function generator is connected to oscilloscope and oscillator via 2 cables.	1
One end of wire is connected to the oscillator, the other end attached to a load over pulley.	1
Procedures	[max 4]
Measure mass m of load with electronic balance. Measure length L of wire with metre rule. Mass per unit length: $\mu = \frac{m}{L}$	1
Adjust the frequency of function generator until a stationary wave of two loops with large amplitude is formed on wire.	1
Method of determining period T from oscilloscope screen.	1
Frequency: $f = \frac{1}{T}$	1
Control of Variables	[max 2]
Tension T in wire kept constant by using the same load.	1
Length L of wire kept constant by measuring separation between oscillator and pulley.	1
Same mode of vibration.	1
Reliability	[max 1]
To determine period T accurately, adjust time-base of CRO to produce a large waveform or use multiple complete waveforms.	1
To accurately determine frequency f of 2 nd harmonic, repeat step 4. by lowering the frequency from a high value. Calculate the average frequency.	1
Analysis	[max 2]
Linearise the equation and state the variables to be plotted.	1
State that n is found from the gradient of the graph.	1
Safety Precautions (not asked in question, but marks awarded)	[max 1]
Wear protective eyewear to prevent injuries to eyes, in case the wire snaps.	1
Wear protective footwear to prevent injuries to the feet, in case the load drops.	1
Secure the oscillator and retort stand to the bench with G-clamps to prevent them from toppling and causing injuries.	1

[2021/P4/Q3(g)] – continuation of WS10

The behaviour of the oscillating system depends on the length of the bottom plank.

It is suggested that the period T is directly proportional to the length L of the plank.

Explain how you would investigate this relationship.

Your account should include:

- your experimental procedure
- control of variables
- how you would use your results to show direct proportionality
- why you might have difficulties using planks with very small lengths and very large lengths.

1. Set up the apparatus as shown in Fig. 1.1 (refer to WS10).

2. Measure the length L of the bottom plank using a meter rule.

3. Repeat step (b) (refer to WS10) and measure the time taken t for N number of oscillations with a stopwatch such that $t > 20$ s. Calculate period $T = t/N$.

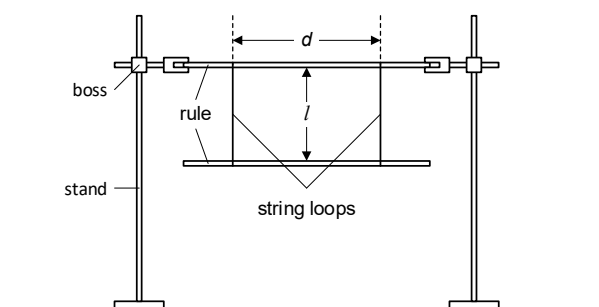
4. Repeat steps 2 and 3 for a total of 8 sets of readings of L and T by using bottom planks of different lengths.

5. Measure the distances d and l with a meter rule before each set of readings to ensure that they remain constant throughout the experiment.

6. If $T \propto L \Rightarrow T = kL$, where k is a constant. Plot a graph of T against L . If a straight-line graph with zero vertical intercept is obtained, then T is directly proportional to L .

7. Using short planks may produce oscillations that are too fast, and thus increasing the percentage uncertainty in the period. For long planks, the oscillations may be unstable for smaller values of d .

[5]



- (b) Move one end of the bottom rule towards you. Move the other end of the rule away from you. Release the rule. The rule will oscillate about a vertical axis through its centre. Determine the period T of these oscillations.